

A Counterexample to Bidual Stability of Uniform Property (S)

Automated Math Discovery Smoke Test

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Source and Open Question

This packet concerns Open Problem (5) from:

William B. Johnson and Tomasz Kania, *Uniform Property (S)*, arXiv:2602.09106.

The source location is the open-problems section of the paper.

(5) If X has uniform property (S), must X^{**} have it too? The converse follows from the principle of local reflexivity.

Problem (5) asks:

If X has uniform property (S), must X^{**} have it too?

We give a negative answer, in the real Banach-space category.

Definitions

For a normed space E , the paper defines

$$U_E(x, y; a) = \sup_{\|z\| \leq a} | \|x + z\| - \|y + z\| |$$

for $x, y \in S_E$ and $a > 0$, and

$$U_E(d; a) = \inf_{\substack{x, y \in S_E \\ \|x - y\| \geq d}} U_E(x, y; a).$$

The space E has uniform property (S) precisely when $U_E(d; a) > 0$ for every $0 < d < 2$ and every $a > 0$.

Counterexample

Theorem 1. *There is a Banach space X with uniform property (S) such that X^{**} fails uniform property (S).*

Proof. Let

$$X = L_1[0, 1]$$

over the real scalars. By Theorem 1.2 of the source paper, $L_1(\mu)$ has uniform property (S) whenever μ is atomless. Thus X has uniform property (S).

We now show that X^{**} does not. Since $X^* = L_\infty[0, 1]$, we have

$$X^{**} = (L_\infty[0, 1])^*.$$

Represent the commutative unital Banach lattice algebra $L_\infty[0, 1]$ as $C(K)$ for its Stone/Gelfand compact space K . The space K has at least two distinct points: for instance, the non-trivial idempotent $\chi_{[0, 1/2]}$ gives a non-trivial clopen set in the Stone representation. Hence

$$X^{**} \cong C(K)^* = M(K),$$

the Banach lattice of regular signed measures on K with the total-variation norm, and there exist two distinct Dirac measures δ_s, δ_t in $M(K)$.

Define two positive norm-one measures

$$u = \frac{3}{4}\delta_s + \frac{1}{4}\delta_t, \quad v = \frac{1}{4}\delta_s + \frac{3}{4}\delta_t.$$

Then

$$\|u - v\| = \left| \frac{3}{4} - \frac{1}{4} \right| + \left| \frac{1}{4} - \frac{3}{4} \right| = 1.$$

We claim that

$$U_{X^{**}}(u, v; 1/8) = 0.$$

Let $z \in M(K)$ satisfy $\|z\| \leq 1/8$. Write

$$z = \alpha\delta_s + \beta\delta_t + z_0,$$

where z_0 has no mass at s or t . Then $|\alpha| \leq 1/8$ and $|\beta| \leq 1/8$. In particular, all four numbers

$$\frac{3}{4} + \alpha, \quad \frac{1}{4} + \beta, \quad \frac{1}{4} + \alpha, \quad \frac{3}{4} + \beta$$

are positive. Since the atoms $\{s\}$, $\{t\}$, and $K \setminus \{s, t\}$ are disjoint for the variation norm,

$$\begin{aligned} \|u + z\| &= \left(\frac{3}{4} + \alpha \right) + \left(\frac{1}{4} + \beta \right) + \|z_0\| \\ &= 1 + \alpha + \beta + \|z_0\|, \end{aligned}$$

and similarly

$$\begin{aligned} \|v + z\| &= \left(\frac{1}{4} + \alpha \right) + \left(\frac{3}{4} + \beta \right) + \|z_0\| \\ &= 1 + \alpha + \beta + \|z_0\|. \end{aligned}$$

Thus $\|u + z\| = \|v + z\|$ for every z with $\|z\| \leq 1/8$. Consequently,

$$U_{X^{**}}(u, v; 1/8) = 0.$$

Since $u, v \in S_{X^{**}}$ and $\|u - v\| = 1$, this implies

$$U_{X^{**}}(1; 1/8) = 0.$$

Therefore X^{**} fails uniform property (S). □

Verification Notes

The proof is exact. The script `code/check_atomic_band.py` samples perturbations in the finite-dimensional model $\ell_1^2 \oplus_1 \mathbb{R}^k$ and checks the identity

$$\|u + z\|_1 = \|v + z\|_1 \quad (\|z\|_1 \leq 1/8)$$

for the same atom coefficients. This is only a sanity check; the measure-theoretic calculation above is the certificate.

Limitations and Novelty Check

This packet answers Problem (5) as stated for real Banach spaces. The example is deliberately classical: $X = L_1[0, 1]$ is one of the paper’s basic positive examples, and the failure in the bidual is caused by two Dirac atoms in the measure representation of $(L_\infty[0, 1])^*$.

I have not performed a literature search beyond the source paper and the local pipeline corpus, so the novelty status is “candidate” pending human and literature review.